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Case Scenario : MySQL Queries Implementation for College Admission Form

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mysql>
mysql> -- EXPERIMENT 24: Case Scenario - College Admission Form
mysql> DROP DATABASE IF EXISTS college_db;
Query OK, 3 rows affected (0.05 sec)

mysql> CREATE DATABASE college_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE college_db;
Database changed
mysql>
mysql> CREATE TABLE Applicant (
->   app_id INT PRIMARY KEY,
->   name VARCHAR(50) NOT NULL,
->   dob DATE,
->   gender VARCHAR(10),
->   email VARCHAR(100) UNIQUE,
->   phone VARCHAR(15)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE Course (
->   course_id INT PRIMARY KEY,
->   course_name VARCHAR(50) NOT NULL,
->   seats_available INT NOT NULL
-> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE Admission (
->   admission_id INT AUTO_INCREMENT PRIMARY KEY,
->   app_id INT,
->   course_id INT,
->   admission_date DATE NOT NULL,
->   status VARCHAR(20) DEFAULT 'PENDING',
->   FOREIGN KEY (app_id) REFERENCES Applicant(app_id),
->   FOREIGN KEY (course_id) REFERENCES Course(course_id)
-> );
Query OK, 0 rows affected (0.04 sec)
```

```

mysql>
mysql> INSERT INTO Applicant VALUES
  -> (1,'Sanjay Kumar','2007-06-12','Male','sanjay@mail.com','9990001111'),
  -> (2,'Anitha Raj','2007-03-25','Female','anitha@mail.com','9990002222'),
  -> (3,'Vignesh Babu','2006-11-05','Male','vignesh@mail.com','9990003333'),
  -> (4,'Keerthana S','2007-09-18','Female','keerthana@mail.com','9990004444');
Query OK, 4 rows affected (0.00 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO Course VALUES
  -> (1,'B.E. Computer Science',60),
  -> (2,'B.E. Electronics',60),
  -> (3,'B.Tech IT',40);
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO Admission (app_id, course_id, admission_date, status) VALUES
  -> (1,1,'2026-06-01','APPROVED'),
  -> (2,1,'2026-06-02','PENDING'),
  -> (3,2,'2026-06-02','APPROVED');
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql>
mysql> SELECT a.name, c.course_name, ad.admission_date, ad.status
  -> FROM Admission ad
  -> JOIN Applicant a ON ad.app_id = a.app_id
  -> JOIN Course c ON ad.course_id = c.course_id;
+-----+-----+-----+-----+
| name      | course_name      | admission_date | status  |
+-----+-----+-----+-----+
| Sanjay Kumar | B.E. Computer Science | 2026-06-01    | APPROVED |
| Anitha Raj   | B.E. Computer Science | 2026-06-02    | PENDING  |
| Vignesh Babu | B.E. Electronics    | 2026-06-02    | APPROVED |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

```

```
mysql>
mysql> SELECT c.course_name, COUNT(*) AS applicant_count
-> FROM Admission ad JOIN Course c ON ad.course_id = c.course_id
-> GROUP BY c.course_name;
```

course_name	applicant_count
B.E. Computer Science	2
B.E. Electronics	1

2 rows in set (0.00 sec)

```
mysql>
mysql> SELECT a.name
-> FROM Applicant a
-> LEFT JOIN Admission ad ON a.app_id = ad.app_id
-> WHERE ad.admission_id IS NULL;
```

name
Keerthana S

1 row in set (0.00 sec)

```
mysql>
mysql> UPDATE Admission SET status = 'APPROVED' WHERE app_id = 2;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> SELECT * FROM Admission;
```

admission_id	app_id	course_id	admission_date	status
1	1	1	2026-06-01	APPROVED
2	2	1	2026-06-02	APPROVED
3	3	2	2026-06-02	APPROVED

3 rows in set (0.00 sec)